

Feature Vulnerability-Aware Adversarial Training

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Abstract—Adversarial Training (AT) typically treats all samples uniformly with fixed loss weights and a global perturbation budget, overlooking intrinsic differences in how samples respond to attacks. In this work, we reveal that different samples exhibit significantly distinct *feature vulnerability patterns* under adversarial perturbations. Building on this insight, we propose Feature Vulnerability-Aware Adversarial Training (FVA-AT), which employs a lightweight, independently optimized feature projector to map final-layer features into a low-dimensional embedding space. Sample vulnerability is quantified via (i) the KL divergence between clean and adversarial embeddings and (ii) a feature-space margin. These metrics guide the dynamic adjustment of per-sample loss weights and perturbation budgets during training. Experiments on CIFAR-10, CIFAR-100, and SVHN demonstrate that FVA-AT consistently yields robustness gains—e.g., +4.57% improvement in robust accuracy over PGD-AT on CIFAR-10 with ResNet-18—while effectively mitigating robust overfitting. Moreover, FVA is plug-and-play and seamlessly integrates with existing AT frameworks (e.g., TRADES, MART) to form stronger hybrid defenses. Code is available at <https://github.com/SSonnyboy/FVA-AT>.

Index Terms—adversarial training, feature vulnerability, dynamic weighting, dynamic perturbation budget

I. INTRODUCTION

Deep neural networks (DNNs) have achieved remarkable success across a wide range of applications, yet they remain highly vulnerable to adversarial examples—inputs perturbed by imperceptible but carefully crafted noise that can cause misclassification [1]. In response, *adversarial training* (AT) has emerged as the de facto standard for improving model robustness, effectively minimizing the loss over adversarially perturbed inputs during training [2]. The seminal work of Madry et al. [2] formulates AT as a robust optimization problem and demonstrates strong empirical defenses using multi-step projected gradient descent (PGD) attacks, setting a benchmark for subsequent research.

Despite its effectiveness, conventional AT treats all training samples uniformly—assigning identical loss weights and employing a fixed perturbation budget ϵ for every instance. This one-size-fits-all strategy overlooks the intrinsic heterogeneity among samples in terms of their feature representations and vulnerability to attacks. Recent studies have revealed that such uniform treatment can lead to suboptimal robustness and even exacerbate the well-known robustness-accuracy trade-off [3]. Moreover, robust overfitting—a phenomenon where robust test accuracy degrades after a certain training epoch despite continued training—has been widely observed in standard AT [4], suggesting that not all samples contribute equally to generalization under adversarial settings.

To address these limitations, several adaptive AT methods have been proposed. For instance, GAIAT [5] proposes that hard samples nearer to the decision boundary are more critical but less robust and should be assigned a larger weight. MAIL [6] further proposes three types of probabilistic margins to quantify the closeness for better reweighting adversarial data accordingly. These approaches underscore a growing consensus: *treating samples differently according to their intrinsic properties can significantly enhance robustness*.

In this work, we investigate sample vulnerability from the perspective of the feature space. Through empirical analysis, we observe that during adversarial training, samples exhibit distinctly different vulnerability patterns at the feature level: some maintain stable class-semantic representations even under perturbation, while others suffer severe distortion under even mild attacks. Motivated by this finding, we propose **FVA-AT** (Feature Vulnerability-Aware Adversarial Training), a novel instance-level adaptive framework that dynamically adjusts both the loss weight and perturbation budget for each sample based on the degree of distortion in its semantic feature representation. In contrast to prior approaches that rely on attack success rates or output logit margins, FVA-AT leverages the semantic structure in the representation space to guide adaptation, enabling a more principled and efficient approach to robust learning.

Experiments on CIFAR-10/100 and SVHN show FVA-AT outperforms PGD-AT and mitigates overfitting. It is plug-and-play with TRADES [3] and MART [7]. Results validate *feature-level vulnerability awareness* as a powerful principle.

We summarize the contributions of this paper as follows:

- 1) We observe that different samples exhibit significantly varying degrees of semantic distortion in the feature space under adversarial perturbations. This insight motivates us to design a feature vulnerability-aware sample evaluation metric that quantifies the robustness of each instance in the representation space.
- 2) Building upon this evaluation metric, we propose two key components: *feature vulnerability-aware loss weighting* and *feature vulnerability-aware perturbation budget allocation*. Together, these components constitute our proposed framework, **FVA-AT**, enabling instance-level adaptive control during adversarial training.
- 3) Experimental results using ResNet-18 and WRN models show that the method in this paper can further improve the robustness of neural networks.

II. MOTIVATION

A. Deep features are semantically rich but highly fragile.

Deep neural networks amplify small input perturbations through multiple nonlinear transformations, making deeper layers more sensitive to adversarial attacks. Let $f_{\theta}^{(l)}(x)$ denote the feature representation at layer l , and δ an adversarial perturbation. The sensitivity of layer l is quantified as:

$$\Delta_l = \|f_{\theta}^{(l)}(x + \delta) - f_{\theta}^{(l)}(x)\|_2.$$

Figure 1 visualizes feature distortions across layers, showing that deeper layers exhibit greater vulnerability. Figure 2 further supports this by measuring the Jensen-Shannon (JS) divergence between clean and adversarial samples for 100 CIFAR-10 test images, with Layer 4 showing the highest JS divergence. Thus, our method focuses on the final-layer features as they are most fragile and critical for robustness.

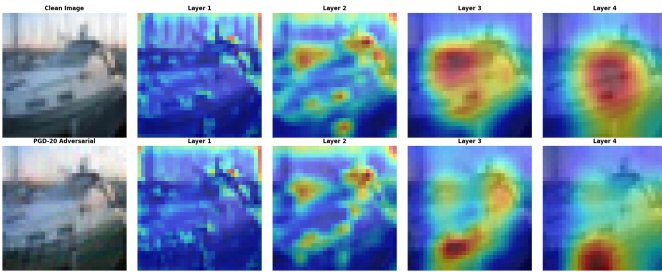


Fig. 1. Visualization of feature attention changes in different layers of ResNet-18. (Top: clean samples; Bottom: adversarial samples)

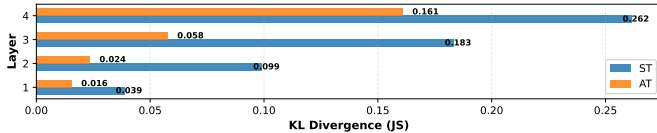


Fig. 2. Average JS divergence across different layers of ResNet-18 for 100 CIFAR-10 test images under two training methods.

B. Samples exhibit distinct feature vulnerability patterns.

Furthermore, we observe that different samples exhibit varying degrees of fragility in their feature representations under adversarial attacks. As shown in the left panel of Figure 3, the Jensen-Shannon (JS) divergence between clean and adversarial samples varies significantly across samples. This indicates that some samples maintain robust feature representations even under adversarial perturbations, while others suffer severe distortions.

To further analyze this phenomenon, we investigate the relationship between feature divergence and attack success rates. Figure 3 demonstrates that the KL divergence of Layer 4 features correlates positively with both loss changes and attack success rates for both standard training (ST) and adversarial training (AT) models. Specifically, the correlation coefficients are $r = 0.531$ for ST and $r = 0.586$ for AT. These findings suggest that the vulnerability of a sample’s final-layer features can effectively predict its susceptibility to adversarial attacks.

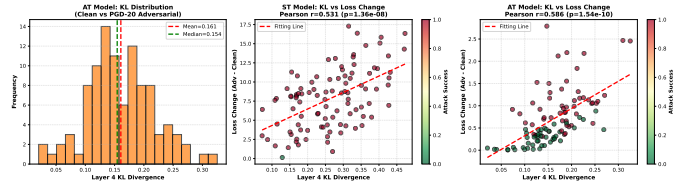


Fig. 3. Relationship between Layer 4 Feature Divergence and Adversarial Attack Success. (Left) KL divergence distribution (AT model) between clean and PGD-20 adversarial examples. (Middle) ST model: KL divergence vs. loss change correlation ($r=0.531$). (Right) AT model: KL divergence vs. loss change correlation ($r=0.586$). Color intensity indicates attack success rate.

C. Adaptive processing based on vulnerability improves robustness.

To investigate the impact of samples with varying feature vulnerability on adversarial training, we perform four operations on each minibatch based on the KL divergence of samples’ final-layer features: **DropLow**: Remove the 20% of samples with the lowest KL divergence. **DropHigh**: Remove the 20% of samples with the highest KL divergence. **BoostHigh**: Set the perturbation budget to 2ϵ for the 20% of samples with the highest KL divergence. **BoostLow**: Set the perturbation budget to 2ϵ for the 20% of samples with the lowest KL divergence.

TABLE I
PERFORMANCE COMPARISON OF DIFFERENT SAMPLE-WISE OPERATIONS ON CIFAR-10 UNDER PGD-10 ATTACKS.

Metric	PGD-AT	DropHigh	DropLow	BoostHigh	BoostLow
Clean	82.48	81.22	82.31	80.35	81.18
PGD-10	52.92	52.17	53.21	52.73	53.49

The experimental results, shown in Table I, demonstrate that **DropLow** and **BoostLow** improve the performance of adversarial training. Specifically, properly focusing on the optimization of highly vulnerable samples and improving the perturbation of highly robust samples can enhance the ability of the model to resist strong attacks. These findings highlight the importance of adaptively adjusting training strategies based on sample-level feature vulnerability. This insight motivates our proposed method, which dynamically adjusts the loss weight and perturbation budget for each sample during adversarial training.

III. METHODOLOGY

Motivated by the empirical observations that samples exhibit varying feature vulnerabilities and that targeted treatment based on feature fragility can improve robustness, we propose Feature Vulnerability-Aware Adversarial Training (FVA-AT). Our approach introduces an independently optimized feature projector that quantifies sample-level feature vulnerability and adaptively adjusts both the loss weight and the perturbation budget for each sample. Figure 4 provides an overview of the proposed method architecture.

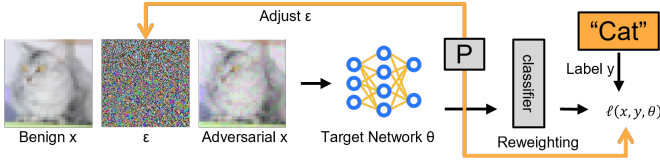


Fig. 4. Overview of FVA-AT. An independently optimized feature projection head adaptively adjusts the loss weight and perturbation budget for each sample based on the final-layer feature map output by the model.

A. Feature Projector

To accurately quantify feature vulnerability during adversarial training, we introduce an independently optimized feature projection layer. Let $f_\theta(x) \in \mathbb{R}^d$ denote the final-layer feature representation output by the base model, where d is the feature dimension (e.g., 512 for ResNet-18). We project this high-dimensional feature into a lower-dimensional embedding space to improve computational efficiency and training stability.

The feature projector consists of a linear transformation followed by ReLU activation and a classifier:

$$\mathbf{z} = \text{ReLU}(\mathbf{W}_1 f_\theta(x) + \mathbf{b}_1), \quad (1)$$

where $\mathbf{W}_1 \in \mathbb{R}^{d_e \times d}$ and $\mathbf{b}_1 \in \mathbb{R}^{d_e}$ are learnable parameters, and d_e is the embedding dimension (default: 128). The embedding $\mathbf{z}$ is then fed into a classification head:

$$\hat{\mathbf{y}}_p = \text{softmax}(\mathbf{W}_2 \mathbf{z} + \mathbf{b}_2), \quad (2)$$

where $\mathbf{W}_2 \in \mathbb{R}^{c \times d_e}$ and $\mathbf{b}_2 \in \mathbb{R}^c$ are classifier parameters, and c is the number of classes.

This projection layer is optimized independently alongside the main model during training, ensuring that the learned embedding effectively captures feature vulnerability patterns. By operating in a lower-dimensional space, we reduce computational overhead while maintaining semantic information necessary for vulnerability assessment.

B. Feature Vulnerability Perception Loss Reweighting

The core insight of our approach is that different samples exhibit varying degrees of feature fragility under adversarial perturbations, and this fragility should be reflected in the training objective. We quantify sample-level feature vulnerability using the KL divergence between clean and adversarial feature embeddings.

For a sample x with its adversarial counterpart $x' = x + \delta$, let $\mathbf{z}_c = f_p(f_\theta(x))$ and $\mathbf{z}_a = f_p(f_\theta(x'))$ denote the projected embeddings from clean and adversarial samples respectively, where f_p is the feature projector and f_θ is the base model. The feature vulnerability is measured as:

$$D_{\text{KL}}(i) = \text{KL}(\text{softmax}(\mathbf{z}_c) \parallel \text{softmax}(\mathbf{z}_a)), \quad (3)$$

where the KL divergence is computed element-wise along the embedding dimension.

Based on this divergence, we compute the reweighting score for each sample by combining feature vulnerability with prediction confidence from the feature projector:

$$w_i = T \cdot \frac{D_{\text{KL}}(i)}{\exp(T \cdot p_i^p)}, \quad (4)$$

where $p_i^p = \max_j \hat{\mathbf{y}}_p^{\text{clean}}[j]$ is the prediction probability for the true class on clean samples as output by the feature projector (Eq. 2), and $T > 0$ is a temperature parameter that controls the sensitivity to projector confidence. This formulation achieves two objectives: (1) samples with higher feature divergence receive larger weights, prioritizing the training of vulnerable samples; (2) the exponential decay term $\exp(-T \cdot p_i^p)$ down-weights confident samples in the feature space, focusing optimization on uncertain ones. By utilizing the feature projector's confidence, we ensure that reweighting directly reflects the model's uncertainty in the projected feature space, strengthening our focus on feature-level vulnerability.

To ensure numerical stability and prevent extreme weight values, we apply normalization after the first 10 training epochs:

$$\hat{w}_i = \frac{\exp(w_i - \max_j w_j) \cdot N}{\sum_k \exp(w_k - \max_j w_j)}, \quad (5)$$

where N is the batch size.

The weighted loss for adversarial training becomes:

$$\mathcal{L}_{\text{robust}} = \frac{1}{N} \sum_{i=1}^N \hat{w}_i \cdot \ell(f_\theta(x'_i), y_i), \quad (6)$$

where ℓ is the base loss function (e.g., cross-entropy for PGD-AT, KL divergence for TRADES, or MART-specific loss). This approach seamlessly integrates with existing adversarial training methods by reweighting their loss contributions at the sample level.

C. Feature Vulnerability Perception ϵ Adjustment

Beyond loss reweighting, we adaptively adjust the per-sample perturbation budget based on feature vulnerability. This is motivated by the observation that robust samples benefit from stronger attacks to expand the decision boundary, while vulnerable samples require moderated perturbations to avoid over-regularization.

We employ a feature-space margin metric to gauge sample-level robustness based on the feature projector's outputs. For adversarially perturbed samples, the feature margin is defined as:

$$m_i^f = \sum_{k \in \text{top-3}} (p_{\text{true}}^p - p_{\text{non-true}}^p(k)), \quad (7)$$

where p_{true}^p is the prediction probability for the true class from the feature projector (Eq. 2) on adversarial samples, and $p_{\text{non-true}}^p(k)$ are the top-3 probabilities for non-target classes from the feature projector. This feature margin quantifies the gap between the true class and competing incorrect classes within the learned feature embedding space, directly reflecting the sample's vulnerability in the feature representation.

TABLE II
COMPARISON OF ROBUSTNESS PERFORMANCE USING RESNET-18 OF 3 RUNS ON CIFAR-10, CIFAR-100, AND SVHN DATASETS. RESULTS ARE REPORTED IN THE FORMAT OF BEST/LAST EPOCH PERFORMANCE. **BOLD** VALUES INDICATE THE BEST PERFORMANCE IN EACH COLUMN.

Methods	Clean	CIFAR-10 PGD-20	AA	Clean	CIFAR-100 PGD-20	AA	Clean	SVHN PGD-20	AA
PGD-AT [2]	82.48 / 83.69	52.05 / 44.71	47.13 / 42.55	56.50 / 56.10	28.68 / 21.04	24.60 / 19.20	88.51 / 87.33	56.07 / 42.74	48.21 / 39.24
TRADES [3]	82.91 / 82.89	52.46 / 50.25	49.23 / 47.19	56.93 / 55.68	29.34 / 27.28	24.32 / 23.71	88.36 / 86.44	56.93 / 48.73	50.87 / 43.56
MART [7]	77.34 / 81.17	56.42 / 51.15	46.67 / 45.04	49.70 / 52.57	30.23 / 25.03	24.78 / 21.61	86.39 / 86.98	58.73 / 51.61	50.28 / 42.02
Cons-AT [8]	82.59 / 83.87	53.67 / 44.68	49.78 / 42.60	57.34 / 56.08	29.47 / 21.38	25.29 / 19.61	88.27 / 86.09	56.71 / 45.77	49.96 / 40.35
RPAT [9]	82.37 / 83.50	52.23 / 44.58	48.20 / 42.48	56.23 / 55.78	29.01 / 21.59	24.67 / 19.62	89.37 / 87.25	55.83 / 42.56	50.25 / 39.00
FVA-AT	80.47 / 80.55	56.62 / 51.43	47.65 / 46.84	55.86 / 54.65	29.33 / 24.86	25.00 / 21.13	86.85 / 86.24	58.59 / 52.29	48.75 / 45.51
FVA-TRADES	81.16 / 79.86	53.54 / 52.69	49.94 / 49.25	58.32 / 53.06	30.62 / 29.11	25.17 / 24.79	88.30 / 83.17	57.10 / 51.15	51.69 / 45.22
FVA-MART	76.83 / 78.90	56.76 / 54.30	48.00 / 47.26	49.79 / 51.84	31.45 / 26.18	25.42 / 22.27	88.83 / 83.29	59.43 / 53.25	49.29 / 43.16

The maximum perturbation budget evolves during training:

$$\epsilon_{\max}(t) = \begin{cases} \epsilon_0 & \text{if } t < t_0 \\ \epsilon_0 + (t - t_0) \cdot \Delta\epsilon & \text{if } t \geq t_0 \end{cases}, \quad (8)$$

where ϵ_0 is the default budget (e.g., 8/255), t_0 is the epoch at which reweighting begins (default: 101), and $\Delta\epsilon = \frac{\epsilon_{\max} - \epsilon_0}{T_{\text{total}} - t_0}$ controls the linear growth rate toward $\epsilon_{\max}$ (e.g., 16/255).

For each sample in a batch, the per-sample perturbation budget is computed as:

$$\epsilon_i = \epsilon_0 + \exp(m_i^f - \max_j m_j^f) \cdot (\epsilon_{\max}(t) - \epsilon_0), \quad (9)$$

where $\exp(m_i^f - \max_j m_j^f)$ applies an exponential transformation to the feature margin with max normalization for numerical stability. Samples with higher feature margins receive exponentially larger scaling factors, allowing them to be trained with larger perturbation budgets. This ensures that the perturbation budget allocation is guided by the model’s confidence in the feature embedding space.

This two-stage approach ensures that: (1) early training focuses on establishing baseline robustness with standard perturbation budgets; (2) later stages enable harder attacks for robust samples to improve generalization to stronger adversaries; (3) vulnerable samples receive adaptive, appropriate perturbation magnitudes based on their current robustness level. The per-sample constraint maintains flexibility while avoiding excessive perturbations that could destabilize training.

IV. EXPERIMENTS

To evaluate the effectiveness of FVA-AT, we conducted extensive experiments on three datasets, including CIFAR10 [10], CIFAR-100 [10] and SVHN [11]. We choose popular ResNet and Wide ResNet as backbones in our experiments. We compare against PGD-AT [2] (ICLR’2018), TRADES [3] (ICML’2019), MART [7] (ICLR’2019), Cons-AT [8] (AAAI’2022), and RPAT [9] (ICCV’2025). The results in Table II are averaged over three independent runs with different random seeds.

A. Experimental Setup

Training Parameters. We train the networks batch stochastic gradient descent with momentum 0.9, weight decay $5e-4$. The training process takes 200 epochs. Batch size is 128, and

the initial learning rate is 0.1. Learning rate is divided by 10 after the 100-th and 150-th epoch. To some extent, this setup can alleviate the impact of adversarial overfitting. All training process is conducted on 6 * Nvidia 3090 GPU.

Hyperparameters. For our proposed FVA-AT, the temperature T is set to 5, the adaptive start-up is enabled after the first decrease in learning rate, which occurs at the 101st epoch, and the maximum perturbation budget ϵ is set to twice the default value, that is, 16/255. **For the detailed ablative studies of these hyperparameters, please refer to Appendix.**

Robustness Evaluation. We evaluated our methods and baselines using the standard accuracy on natural test data and the adversarial robustness based on several representative attack methods under the perturbation distance constraint of $\mathcal{L}_\infty$, including PGD [2], C&W [12] (50 steps), and AutoAttack [13]. For PGD attacks, the maximum iterations are set at 10, 20, and 50, corresponding to PGD-10, PGD-20, and PGD-50, respectively. The perturbation intensity is fixed at 8/255, which is the most extensive evaluation setting.

B. Main Results

Comparing with other methods on three tasks. Table II presents a comprehensive comparison of FVA-AT framework against five baseline adversarial training methods across three benchmark datasets, all evaluated using the ResNet-18 architecture. The FVA-enhanced variant consistently achieves superior robustness, a trend particularly evident in the final-epoch results—which better reflect the model’s generalization capability. On CIFAR-10, FVA-AT yields significant improvements over standard PGD-AT under PGD-20 attacks, with absolute robust accuracy gains of +4.57% and +6.72%. Notably, this advantage generalizes to CIFAR-100 (+0.65%) and SVHN (+2.52%) even without extensive hyperparameter tuning. The consistent performance gains across diverse datasets and attack types suggest that our feature vulnerability-aware approach not only enhances adversarial robustness but also effectively mitigates adversarial overfitting.

Performance when combined with other methods. Moreover, the proposed feature vulnerability-aware (FVA) module is plug-and-play by design and integrates seamlessly with existing strong adversarial training methods to form even more robust defenses. Specifically, on CIFAR-10, FVA-TRADES improves upon TRADES by +1.08% under PGD-20 attack

TABLE III

ROBUSTNESS EVALUATION ON CIFAR-10 USING WIDEResNet-34-10 ARCHITECTURE. ALL RESULTS ARE REPORTED AT THE BEST EPOCH.

Methods	Clean	PGD-10	PGD-20	PGD-50	C&W	AA
PGD-AT [2]	86.19	55.92	54.89	54.71	53.64	51.57
TRADES [3]	84.47	56.78	55.90	55.67	53.63	52.49
MART [7]	80.30	59.60	59.23	59.14	52.82	51.27
Cons-AT [8]	86.51	56.59	55.65	55.47	54.48	52.33
RPAT [9]	85.62	55.90	54.96	54.74	53.68	51.44
FVA-AT	85.74	60.78	59.95	59.84	54.25	51.67
FVA-TRADES	85.01	58.49	57.93	57.75	54.65	53.52
FVA-MART	80.56	61.11	60.77	60.66	53.74	52.10

TABLE IV

COMPARISON WITH OTHER ADAPTIVE AT METHODS ON CIFAR-10 USING RESNET-18. ALL RESULTS ARE REPORTED AT THE BEST EPOCH.

Methods	Clean	PGD-10	PGD-20	PGD-50	C&W	AA
AT	82.48	52.92	52.05	51.96	48.56	47.13
+GAIRAT [5]	81.54	56.79	56.32	56.17	41.67	39.21
+MAIL [6]	82.98	57.32	56.74	56.62	45.85	44.78
+FVA	80.47	57.08	56.62	56.51	49.48	47.65
TRADES	82.91	53.42	52.46	52.24	50.51	49.23
+GAIRAT [5]	82.31	54.57	54.13	53.79	45.23	42.56
+MAIL [6]	81.05	55.65	55.23	55.05	48.42	45.86
+FVA	81.16	54.42	53.54	53.37	50.90	49.94

and by +0.71% under AA. Similarly, FVA-MART achieves a +1.33% robustness gain over MART under AA evaluation.

Performance in other network architectures. To demonstrate the generalizability of our approach across different network architectures, Table III presents results using the wider and deeper WideResNet-34-10 model on CIFAR-10. The results show that FVA-AT consistently improves robustness across all attack variants. FVA-MART achieves the highest PGD accuracies, with 61.11%, 60.77%, and 60.66% on PGD-10, PGD-20, and PGD-50 respectively, demonstrating 1.51%, 1.54%, and 1.52% improvements over vanilla MART. FVA-TRADES delivers the best AutoAttack accuracy of 53.52%, surpassing vanilla TRADES by 1.03% and all other baseline methods. Notably, while the larger model capacity of WideResNet-34-10 provides higher absolute robustness values compared to ResNet-18, our FVA framework continues to provide consistent improvements, validating that feature vulnerability-aware training is architecture-agnostic and can be effectively integrated with different network designs.

Performance comparison with other adaptive methods.

Table IV compares our FVA method against two representative adaptive adversarial training approaches: GAIRAT [5] and MAIL [6]. Although MAIL combined with standard AT achieves strong performance under PGD attacks (56.74% under PGD-20), our FVA-AT attains comparable robustness on PGD (56.62%) while demonstrating significantly superior performance under more challenging attacks—specifically, C&W (49.48% vs. 45.85%) and AutoAttack (47.65% vs. 44.78%). The 2.87% improvement under AutoAttack highlights FVA’s stronger generalization across diverse attack strategies. When integrated with TRADES, FVA further boosts AutoAttack

TABLE V

ABLATION STUDY ON CIFAR-10 WITH RESNET-18. RE- $\mathcal{L}$ DENOTES LOSS REWEIGHTING AND RE- ϵ DENOTES ADAPTIVE EPSILON ADJUSTMENT. “W/O PROJECTOR” USES GROUND-TRUTH LOGITS INSTEAD OF FEATURE PROJECTOR OUTPUTS FOR VULNERABILITY ASSESSMENT.

Re- $\mathcal{L}$	Re- ϵ	Clean		PGD-10		AA	
		best	last	best	last	best	last
PGD-AT		82.48	83.69	52.92	46.30	47.13	42.55
✓		80.72	83.35	56.82	42.37	47.51	41.79
	✓	80.51	80.72	53.27	50.02	47.38	44.62
w/o Projector		82.12	83.56	56.44	48.25	46.74	44.07
✓	✓	80.47	80.55	57.08	52.11	47.65	46.84

accuracy to 49.94%, outperforming plain TRADES by 0.71%, GAIRAT by 7.38%, and MAIL by 4.08%.

A key advantage of FVA over geometry-based methods like GAIRAT is its focus on feature-space vulnerability rather than input-space geometry, yielding a more reliable indicator of sample-wise fragility. Unlike MAIL—which relies on the logit margin from the main network—FVA employs a separately optimized feature projector to adaptively modulate loss weights and perturbation budgets. This design enables more stable training dynamics and consistently better final robustness against strong adaptive attacks.

C. Ablation experiment

Table V presents an ablation study analyzing the contribution of each component in FVA-AT. The baseline PGD-AT achieves 46.30% PGD-10 accuracy and 42.55% AutoAttack accuracy at the last epoch, exhibiting significant adversarial overfitting with a 6.62% and 4.58% drop from the best epoch, respectively. When applying only loss reweighting (Re- $\mathcal{L}$), the best epoch performance improves to 56.82% on PGD-10, but adversarial overfitting remains severe with last epoch accuracy dropping to 42.37%. Conversely, applying only adaptive epsilon adjustment (Re- ϵ) yields 50.02% PGD-10 accuracy at the last epoch, demonstrating that sample-specific perturbation budgets help mitigate overfitting. The combination of both components (Re- $\mathcal{L}$ + Re- ϵ) achieves the best results: 52.11% PGD-10 accuracy and 46.84% AutoAttack accuracy at the last epoch, with significantly reduced overfitting gap (4.97% for PGD-10, compared to 14.45% for Re- $\mathcal{L}$ only).

Beyond the above, we also investigate a key design question closely tied to FVA-AT: *Is a separately optimized feature projector truly necessary?* Could we instead directly use the KL divergence of the main network’s logits to assess sample vulnerability? To address this, we conduct a simple ablation study: we remove the dedicated feature projector and replace the projected embeddings in Eq 3 and the predicted logits in Eq 7 with the raw logits from the main network. Results are reported in Table V.

The “w/o projector” variant—using the main model’s logits directly for vulnerability estimation—achieves 48.25% robust accuracy under PGD-10 and 44.07% under AutoAttack at the final epoch. While this variant still outperforms standard PGD-AT, it lags behind the full FVA-AT by 3.86% (PGD-10) and

2.77% (AutoAttack). This gap validates our design choice of employing an independently optimized feature projector. By operating in a low-dimensional embedding space and being trained with an objective decoupled from the main adversarial task, the projector yields a more stable and effective measure of feature vulnerability.

D. Loss Landscape Visualization

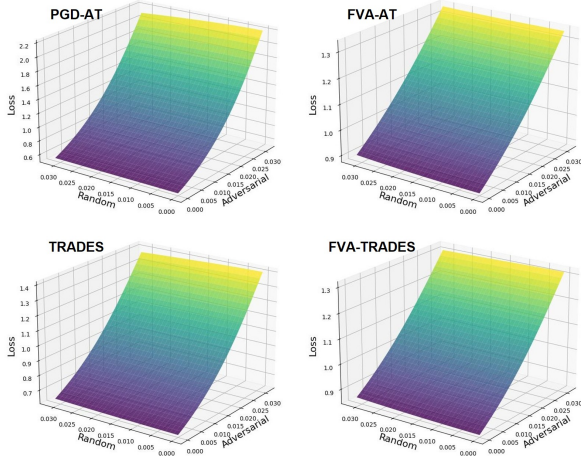


Fig. 5. Loss landscape comparison at the last checkpoints of PGD-AT, FVA-AT, TRADES, and FVA-TRADES method by using 500 CIFAR10 test samples, showcasing variations in cross-entropy loss across a random and the PGD-50 gradient direction.

Figure 5 shows the loss landscape visualizations of different adversarial training methods at the final epoch, including PGD-AT, FVA-AT, TRADES, and FVA-TRADES. We use the first 500 samples from the CIFAR-10 test set to plot the cross-entropy loss over a 2D slice of the input space, constructed from two directions: (i) a random direction and (ii) the gradient direction obtained via PGD-50.

The results suggest that FVA-enhanced methods differ from conventional approaches to some extent. Specifically, PGD-AT and TRADES exhibit less smooth loss surfaces with nonlinear characteristics along the PGD gradient direction, potentially increasing their vulnerability to attacks. In comparison, FVA-AT and FVA-TRADES display more linear loss landscapes with smaller variations in loss magnitude. This indicates that our method better preserves the local linearity of the model around data points, which may contribute to improved adversarial robustness.

V. CONCLUSION

In this work, we uncover that samples exhibit significantly distinct feature vulnerability patterns under adversarial perturbations—some preserve semantic structure in deep features, while others suffer severe distortion. Based on this, we propose **Feature Vulnerability-Aware AT (FVA-AT)**, employing a lightweight projector to quantify vulnerability via KL divergence and margins, driving dynamic loss reweighting

and budget allocation. Extensive experiments show that FVA-AT consistently improves robust accuracy (e.g., +4.57% over PGD-AT on CIFAR-10), effectively mitigates robust overfitting, and generalizes across datasets (CIFAR-100, SVHN) and architectures (ResNet, WideResNet) without additional tuning. Moreover, thanks to its plug-and-play nature, FVA can seamlessly enhance existing AT frameworks such as TRADES and MART, offering a practical and effective pathway toward more adaptive and robust deep learning.

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